

Module 2 Cheat Sheet – Introduction to Linux Commands

Getting Information

Return your user name:

None
`whoami`

Return your user and group ID:

None
`id`

Return operating system name, username, and other info:

None
`uname -a`

Display reference manual for a command:

None
`man top`

List available `man` pages with descriptions:

None
`man -k .`

Get help on any command (e.g., curl):

None

```
curl --help
```

This provides a brief overview of the command's usage and options.

Return the current date and time:

None

```
date
```

Navigating and Working with Directories

List files and directories by date (newest last):

None

```
ls -lrt
```

Find files in directory tree that end in `.sh`:

None

```
find -name "*.sh"
```

Return path to present working directory:

None

```
pwd
```

Create a new directory:

None

```
mkdir new_folder
```

Change current directory:

- Up one level:

None

```
cd ../
```

- To home:

None

```
cd ~
```

or

None

```
cd
```

- To another directory:

None

```
cd path_to_directory
```

Remove directory (verbose):

None

```
rmdir temp_directory -v
```

Monitoring System Performance and Status

List running processes:

None

`ps`

List all processes:

None

`ps -e`

Display resource usage:

None

`top`

List mounted file systems and usage:

None

`df`

Creating, Copying, Moving, and Deleting Files

Create or update a file:

None

`touch a_new_file.txt`

Copy a file:

None

```
cp file.txt new_path/new_name.txt
```

Move or rename a file:

None

```
mv this_file.txt that_path/that_file.txt
```

Remove a file (verbose):

None

```
rm this_old_file.txt -v
```

Working with File Permissions

Change/modify file permissions to 'execute' for all users:

None

```
chmod +x my_script.sh
```

Change/modify file permissions to 'execute' only for you, the current user:

None

```
chmod u+x my_file.txt
```

Remove 'read' permissions from group and other users:

None

```
chmod go-r
```

Displaying File and String Contents

Display file contents:

```
None  
cat my_shell_script.sh
```

Display file page-by-page:

```
None  
more ReadMe.txt
```

Show first 10 lines:

```
None  
head -10 data_table.csv
```

Show last 10 lines:

```
None  
tail -10 data_table.csv
```

Display text or variables:

```
None  
echo "I am not a robot"  
echo "I am $USERNAME"
```

Basic Text Wrangling

Sorting and Removing Duplicates

Sort and display lines of file alphanumerically:

None

```
sort text_file.txt
```

Sort in reverse:

None

```
sort -r text_file.txt
```

Drop consecutive duplicated lines and display result:

None

```
uniq list_with_duplicated_lines.txt
```

Displaying basic stats:

Count lines:

None

```
wc -l table_of_data.csv
```

Count words:

None

```
wc -w my_essay.txt
```

Count characters:

None

```
wc -m some_document.txt
```

Extracting lines of text containing a pattern:

Some frequently used options for **grep**

Common options:

Option	Description
-n	Print line numbers along with matching lines
-c	Get the count of matching lines
-i	Ignore the case of the text while matching
-v	Print all lines which do not contain the pattern
-w	Match only if the pattern matches whole words

Extract lines containing the word "hello", case insensitive and whole words only:

None

```
grep -iw hello a_bunch_of_hellos.txt
```

Extract lines containing the pattern "hello" from all files in the current directory ending in .txt:

None

```
grep -l hello *.txt
```

Merge two or more files line-by-line, aligned as columns:

Suppose you have three files containing the first and last names of your customers, plus their phone numbers.

Use **paste** to align file contents into a Tab-delimited table, one row for each customer:

None

```
paste first_name.txt last_name.txt phone_number.txt
```


Use comma delimiter:

None

```
paste -d "," first_name.txt last_name.txt phone_number.txt
```

Use the `cut` command to extract a column from a table-like file:

Suppose you have a text file whose rows consist of first and last names of customers, delimited by a comma.

Extract first names, line-by-line:

None

```
cut -d "," -f 1 names.csv
```

Extract the second to fifth characters (bytes) from each line of a file:

None

```
cut -b 2-5 my_text_file.txt
```

Extract the characters (bytes) from each line of a file, starting from the 10th byte to the end of the line:

None

```
cut -b 10- my_text_file.txt
```

Compression and Archiving

Archive a set of files:

None

```
tar -cvf my_archive.tar.gz file1 file2 file3
```

Compress files:

None

```
zip my_zipped_files.zip file1 file2
zip my_zipped_folders.zip directory1 directory2
```

Extract files from a compressed zip archive:

None

```
unzip my_zipped_file.zip
unzip my_zipped_file.zip -d extract_to_this_directory
```

Working with networking commands

Print hostname:

None

```
hostname
```

Send packets to URL and print response:

None

```
ping www.google.com
```

Display or configure system network interfaces:

None

```
ip
```

Display contents of file at a URL:

None

```
curl <url>
```

Download file from a URL:

None

```
wget <url>
```
